Minh-Nhat Nguyen

Trier, Germany · nnguyennhat201@gmail.com · GitHub · LinkedIn · Google Scholar

RESEARCH INTERESTS

Vision-language models can be confidently wrong — my research closes that gap. I build inference-time self-correction mechanisms, diagnose failure modes such as hallucination, and design benchmarks that measure epistemic reliability rather than raw accuracy, moving multimodal systems toward genuine reliability, not just performance.

EDUCATION

University of Trier, Germany — Oct 2025 — Present
M.Sc. in Data Science

University of Economics Ho Chi Minh City (UEH), Vietnam — 2020 — 2024
B.Sc. in Data Science, Honors — Final GPA: 3.5/4.0

Relevant coursework: Data Mining, Machine Learning, Artificial Intelligence, Natural Language Processing, Data Visualization, Mathematical Statistics, Econometrics.

AI Vietnam (AIVN) — AIO 2023 Program — 2023
Intensive training in mathematics for AI, machine learning, multimodal deep learning, generative models, and advanced architectures.

PUBLICATIONS

Nguyen, T.-H., Nguyen, M.-N., Nguyen, N.-H., et al. (2026). ESC: Emotional Self-Correction for Reliable Vision-Language Models. To appear in Proceedings of the European Conference on Computer Vision (ECCV 2026). **Co-first author / equal contribution.**

Nguyen-Nhu, T.-A., Tran, H.-L., Le, N.-K., Nguyen, M.-N., et al. (2025). A Lightweight Moment Retrieval System with Global Re-Ranking and Robust Adaptive Bidirectional Temporal Search. Presented at CVPRW 2025.

RESEARCH EXPERIENCE

GenAI4E Lab — ECCV 2026
Research Contributor — ESC: Emotional Self-Correction for Reliable Vision-Language Models

- Designed the theoretical framework for emotional self-correction, grounding the approach in Russell's Circumplex Model of Affect.
- Built an extensible codebase supporting multiple target and verifier VLMs across four benchmark families.
- Conducted experiments spanning safety, hallucination, reasoning, and vision-centric perception benchmarks.
- Designed and ran controlled rebuttal ablations isolating emotion as the causal factor behind performance gains.

TECHNICAL SKILLS

Languages & Tools: Python, SQL (PostgreSQL), Bash, Git.

Frameworks: PyTorch, TensorFlow, Pandas, NumPy, Scikit-Learn.

Focus Areas: Vision-Language Models, Computer Vision, NLP, Data Mining.

CERTIFICATIONS & LANGUAGES

IBM Data Science Professional Certificate; Machine Learning Specialization (Coursera); HackerRank SQL (Intermediate).

Vietnamese (native); English (IELTS 7.5); German (intermediate, in progress).